Streamlining IT Procurement: Automating Standard Laptop Orders with WI

A Project Report on Workflow-based Automation for IT Asset Procurement

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1. Abstract

In most organizations, ordering standard laptops for new joiners or replacements is still done through emails, spreadsheets and long approval chains. This creates delay, missing records and extra work for the IT and procurement teams. This project presents a simple automated solution using Workflow Intelligence (WI) to handle the complete laptop order process. The system starts when an employee or HR raises a procurement request. It then checks the available budget, routes the request for manager approval, selects the approved vendor, updates inventory and finally places the order. The goal is to reduce manual effort, improve transparency and make the whole process faster and more reliable. This report explains the problem, the current way of working, the proposed automated system, its architecture, benefits and future improvements.

2. Introduction

Every company needs laptops for its employees. When a new employee joins or an old laptop is damaged, the IT team must order a new machine. In many places this process is still manual. A request is raised by email or in a shared Excel sheet. The manager has to approve it, finance has to check the budget, and procurement has to choose the vendor and place the order. Because many people are involved and there is no single system, requests get delayed or lost.

This project focuses on automating the standard laptop ordering process with the help of Workflow Intelligence (WI). WI is used to design and run a clear, step-by-step automated workflow. The system takes care of the procurement request, manager approval, budget check, vendor selection, inventory update and order placement. By connecting these steps in one flow, the organization can reduce waiting time and keep better records of every laptop ordered.

3. Problem Statement

The present method of ordering standard laptops has several problems:

- Employees raise requests through email or paper forms, so there is no single place to track status.
- Manager approval depends on the person being free and checking email regularly, which causes delay.
- Budget is often checked only at the end, so sometimes the request is rejected after many days of waiting.
- Vendor selection is done manually and may not always follow the approved list of suppliers.
- Inventory records are updated late or not at all, so the company does not know how many laptops are available or on order.
- There is no clear history of who requested, who approved and when the order was placed.

Because of these issues, the IT procurement process becomes slow, error-prone and hard to audit. A structured automated workflow is needed to solve these problems.

4. Objectives

The main objectives of this project are:

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1	To design an automated workflow for raising and tracking standard laptop procurement requests.
2	To include automatic manager approval and budget check steps inside the same workflow.
3	To select the correct vendor from a pre-approved list based on laptop model and price.
4	To update inventory records as soon as an order is confirmed.
5	To reduce the total time taken from request to order placement and keep a clear audit trail.

5. Existing System

In the current system, the laptop ordering process works roughly as follows:

An employee or HR team member sends an email or fills a paper form with the required laptop details (model, quantity, department). The request is then forwarded to the reporting manager. After the manager replies with approval, the request goes to the finance or IT admin team for budget check. Only after budget confirmation does the procurement team look for a vendor, compare prices and place the purchase order. Finally someone updates a shared Excel sheet that acts as the inventory list.

This approach has no central system. Status of a request is known only by asking people. There is high chance of duplicate requests, wrong laptop models being ordered, or budget being exceeded. The process can take several days or even weeks depending on how quickly each person responds.

6. Proposed System

The proposed system uses Workflow Intelligence (WI) to create one continuous automated flow for standard laptop orders. All steps are connected so that the request moves forward automatically when conditions are met.

Key features of the proposed system:

- A simple web form or portal where the requester enters laptop model, quantity, employee details and department.
- Automatic creation of a procurement request with a unique request ID.
- Built-in manager approval step that sends notification and waits for approve / reject decision.
- Automatic budget check against the department budget allocated for IT assets.
- Vendor selection logic that picks the preferred supplier for that laptop model from an approved list.
- Automatic update of inventory (available stock and items on order).
- Email or dashboard notifications at every important stage so the requester and IT team can see the status.
- Complete history of every action for audit and reporting.

The system is designed only for standard laptop models that are already approved by the company. Special or high-cost machines can still follow a separate manual process if needed.

7. System Architecture & Workflow

The architecture is a linear automated workflow managed by Workflow Intelligence (WI). Each step has a clear input, decision point and output. The main components are: Request Form, Approval Engine, Budget Service, Vendor Catalogue, Inventory Module and Notification Service.

Step-by-step laptop order automation:

Step 1 – Raise Procurement Request

The employee or HR opens the laptop request form. They select the standard laptop model from a drop-down, enter quantity, employee ID and department. On submit, WI creates a new procurement request with a unique ID and status “Pending Manager Approval”.

Step 2 – Manager Approval

WI sends an approval task (email or dashboard notification) to the reporting manager. The manager can Approve or Reject with comments. If rejected, the requester is informed and the flow ends. If approved, the request moves to the next step.

Step 3 – Budget Check

The system looks up the remaining IT budget for that department. If the cost of the requested laptops is within the available budget, the request is marked “Budget Approved”. If not, it is sent back with a message and the flow stops or waits for additional budget allocation.

Step 4 – Vendor Selection

WI reads the approved vendor list for the selected laptop model. It chooses the preferred vendor based on price, delivery time or company preference rules. The selected vendor and expected price are recorded on the request.

Step 5 – Inventory Update

Before placing the order, the system checks current stock. If stock is already available, it can reserve the laptop and mark the request as “Fulfilled from Inventory”. Otherwise it creates an “On Order” entry in the inventory module so that stock levels stay accurate.

Step 6 – Place Order & Notify

WI generates the purchase order data and either sends it to the vendor system or creates a task for the procurement team to place the order. Status is updated to “Order Placed”. The requester, manager and IT team receive a final notification with the expected delivery date.

The entire flow is designed so that most standard requests complete without any manual intervention after the manager has given approval. Only exceptions (budget shortage, special models, vendor issues) need human attention.

8. Benefits

Benefit Area	Description
Faster Processing	Average time from request to order placement is reduced because waiting for emails is replaced by automatic routing.
Better Visibility	Requester and IT team can see the live status of every request in one place.
Budget Control	Budget is checked early, so overspending is avoided before the order is placed.
Correct Vendor	Only approved vendors are used, which helps in price consistency and compliance.
Accurate Inventory	Stock and on-order quantities are updated automatically, reducing missing or duplicate records.
Audit Trail	Every approval, budget decision and order action is logged for future review.
Less Manual Work	IT and procurement staff spend less time chasing emails and more time on exceptions and planning.

9. Conclusion

This project shows that the standard laptop ordering process can be made much smoother by using Workflow Intelligence (WI). By connecting the procurement request, manager approval, budget check, vendor selection and inventory update into one automated flow, the organization can save time, reduce errors and keep better control of IT spending. The proposed system is simple enough for a final-year project yet practical enough to be used in a real company for everyday laptop requests. It removes most of the

manual follow-up work and gives clear visibility to everyone involved.

10. Future Scope

The current design focuses only on standard laptop models. In future the system can be extended in the following ways:

- Support for other IT assets such as monitors, docking stations and mobile phones using the same workflow pattern.
- Integration with the company's ERP or finance system so that purchase orders are created automatically.
- Mobile-friendly request and approval screens so managers can approve from their phone.
- Dashboard with charts showing average approval time, budget usage and most ordered laptop models.
- Machine learning suggestions that recommend the best laptop model based on role and past orders.
- Multi-level approval for high-value or non-standard requests.

11. References

1. Workflow Automation Concepts – Introduction to Business Process Management, academic notes on process automation.

2. IT Asset Management Best Practices – Guidelines for tracking hardware lifecycle and procurement.

3. Approval Workflow Design – Common patterns for manager approval and budget validation in enterprise systems.

4. Vendor Management in Procurement – Principles of maintaining approved supplier lists and selection rules.

5. Inventory Control Basics – Methods for updating stock levels and on-order quantities in real time.

6. SmartBridge / Skill Wallet project guidelines and template structure used for report formatting only.